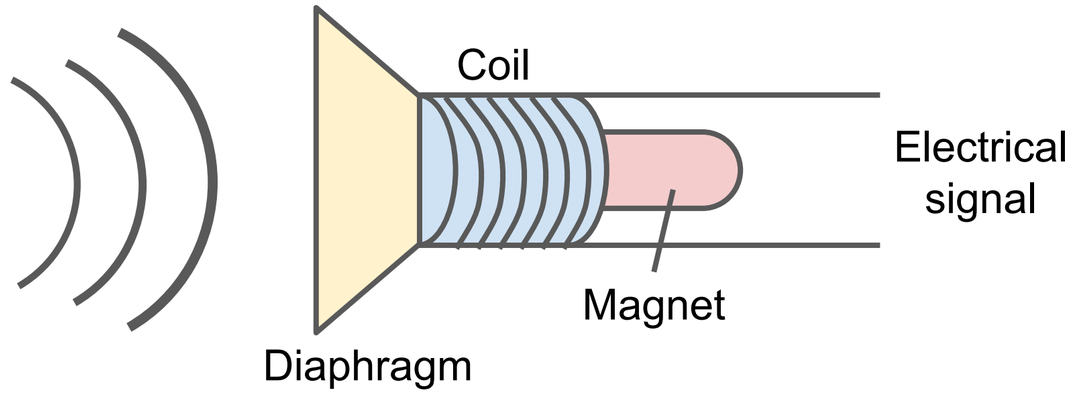
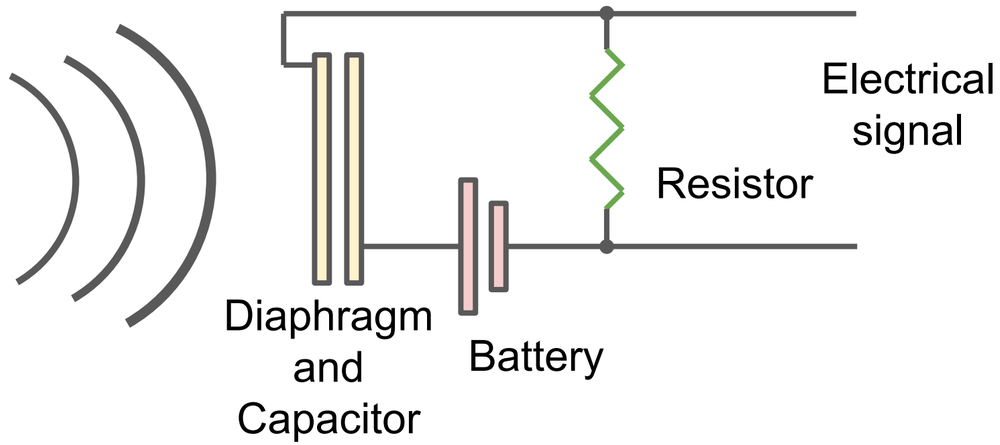
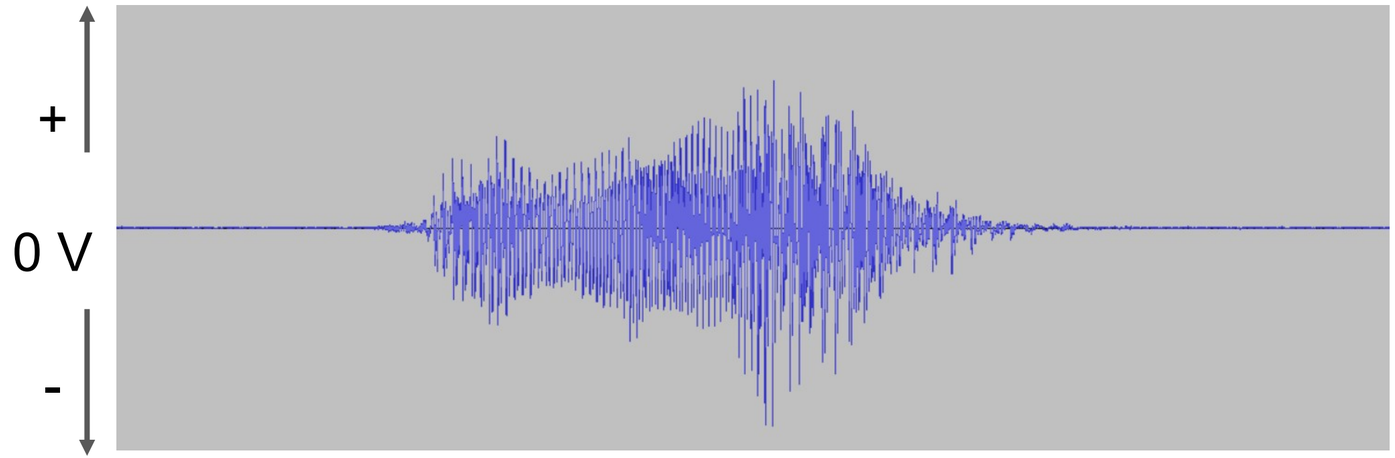
## Dynamic Microphone




## Condenser Microphone

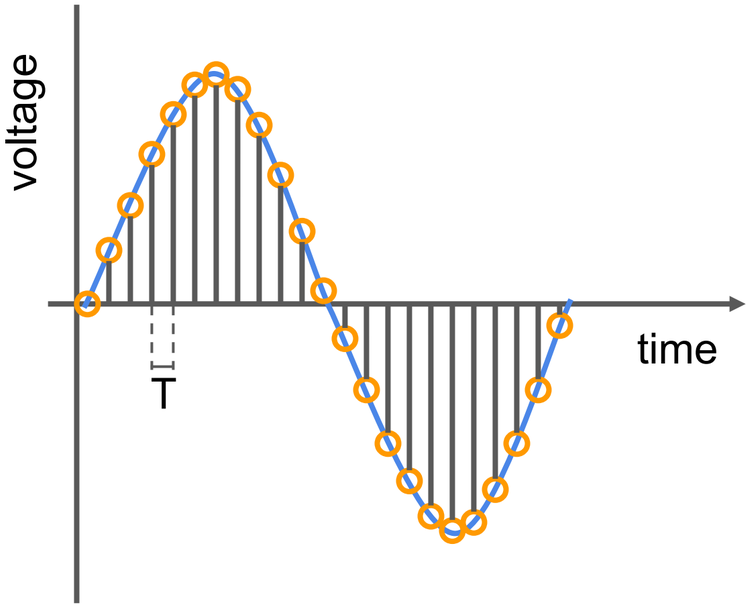



## “hello”




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# Sampling Rate

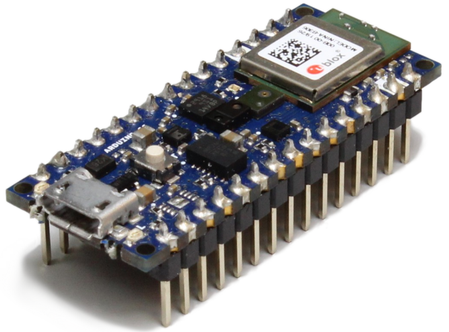



Sampling period (T): 62.5 µs
Sampling rate ($f_s$): 1/62.5 µs = 16 kHz



Data collection:
$f_s$ = 44.1 kHz

To training

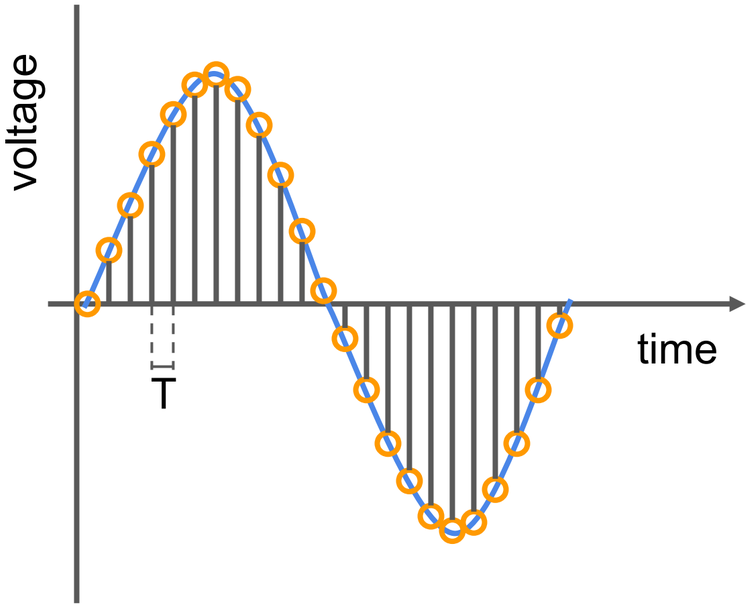


Inference:
$f_s$ = 16 kHz

Model

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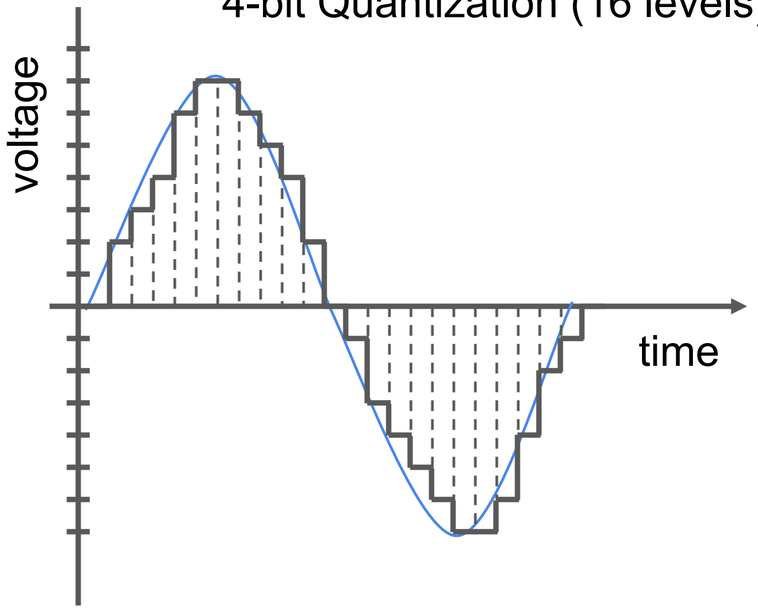
# Bit Depth



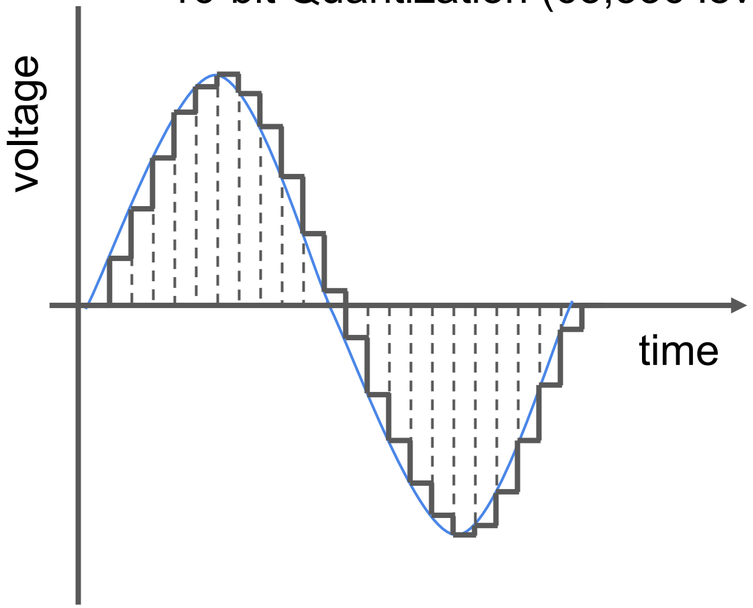

Sampling period (T): 62.5 µs

Sampling rate ($f_s$): 1/62.5 µs = 16 kHz

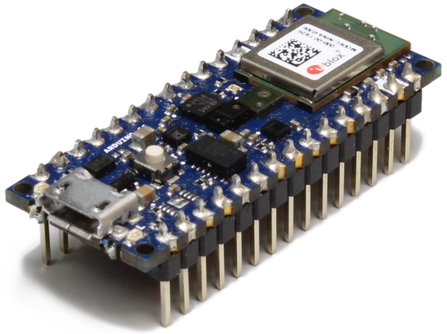
4-bit Quantization (16 levels)

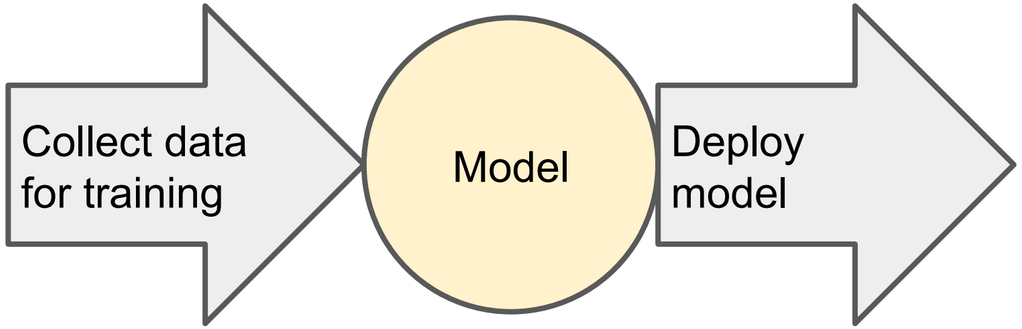


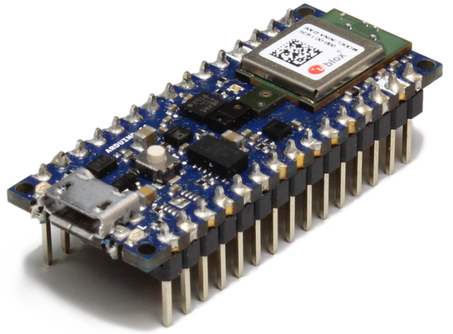

16-bit Quantization (65,536 levels)




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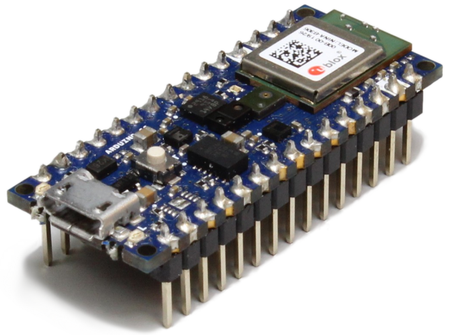
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Collect data for training

Model

Deploy model



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